Grade 6
Unit 7
Lesson 2 Practice

Name______Answer Key_____
Date______________________
Period_____________________

Find the prime factorization of each value.

1. 275

5 55

5 11

$5 \cdot 5 \cdot 11$ or $5^2 \cdot 11$

2. 160

2 80

2 40

2 20

2 10

2 5

$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5$ or $2^5 \cdot 5$

3. 232

2 116

2 58

2 29

$2 \cdot 2 \cdot 2 \cdot 29$ or $2^3 \cdot 29$

4. 54

2 27

3 9

3 3

$2 \cdot 3 \cdot 3 \cdot 3$ or $2 \cdot 3^3$

5. 2550

5 510

5 102

2 51

3 17

$2 \cdot 3 \cdot 5 \cdot 5 \cdot 17$ or $2 \cdot 3 \cdot 5^2 \cdot 17$

6. 1020

2 510

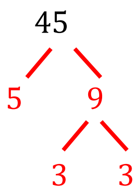
5 102

2 51

3 17

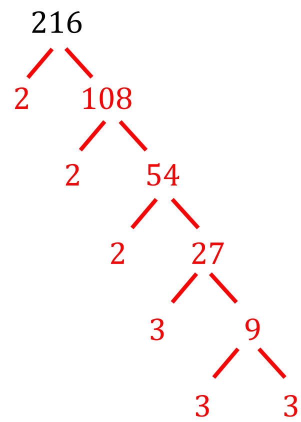
$2 \cdot 2 \cdot 3 \cdot 5 \cdot 17$ or $2^2 \cdot 3 \cdot 5 \cdot 17$

7.



$3 \cdot 3 \cdot 5$ or $3^2 \cdot 5$

8.



$2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3$ or $2^3 \cdot 3^3$

9. Find a number that meets the following conditions.

- The value of the number is less than 500.
- One of its prime factors has an exponent of 5.
- When expressed in exponential form, its prime factorization has 3 bases.

$2^5 \cdot 3 \cdot 5 = 480$

10. Find a number that meets the following conditions.

- The value of the number is less than 5000.
- One of its prime factors has an exponent of 2.
- When expressed in exponential form, its prime factorization has 5 bases.

$2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 4620$